



Side-chain fluorinated polymer surfactants in biosolids from wastewater treatment plants

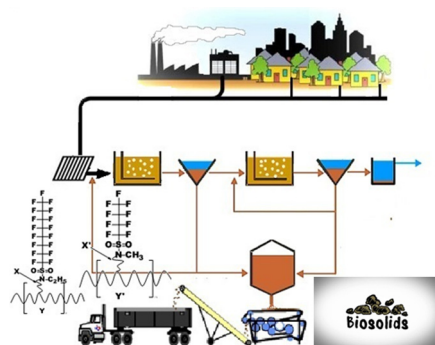
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GRAPHICAL ABSTRACT



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ABSTRACT

High concentrations of the main components in Scotchgard[™] fabric protector products (pre-2002 and post-2002; side-chain fluorinated polymer surfactants, S1 and S2, respectively) were detected in biosolids samples from twenty pan-Canadian wastewater treatment plants (WWTPs). Based on mass spectrometric analysis, S1 and S2 can be named as side-chain perfluorooctane sulfonamide-urethane polymer and side-chain perfluorobutane sulfonamide-urethane polymer, respectively. S1 (with C₈F₁₇ side-chain) concentrations ranged from 1.08–105 ng/g d.w. and S2 (with C₄F₉ side-chain) concentrations ranged from 37.5–2051 ng/g d.w., which were much higher than that of other commonly monitored perfluoroalkyl substances (PFAS). S1 and S2 concentrations were significantly correlated ($p < 0.001$; $r^2 = 0.6142$) indicating similar source origins. A negative linear correlation was observed ($p < 0.05$) between concentrations of S1 (or S2) with the volume of WWTP treated wastewater per day per person (m³/person/day). The total concentration of 22 other PFAS ranged from 4.93 to 92.6 ng/g d.w., and approximately thirty times lower than S1 and S2 concentrations. The calculated elemental fluorine concentrations of ΣF_{S1&S2} were generally much higher than the sum of the other PFAS. PFAS concentrations in biosolids are likely underestimated without consideration of S1 and S2.

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1. Introduction

Poly- and per-fluoroalkyl substances (PFAS) are a large class of fluorinated organic chemicals, which in most cases contain a fluorinated hydrophobic alkyl chain attached to a hydrophilic moiety (Buck et al., 2011; Prevedouros et al., 2006). For over 50 years, PFAS have been widely used as fire retardants, oil and water repellents for e.g. in furniture and clothing water-proofing (Ahrens, 2011; Arvaniti and Stasinakis, 2015; Schultes et al., 2018; Ateia et al., 2019). However, as exemplified by perfluorooctane sulfonate (PFOS), PFAS in recent years have been under increasing scrutiny and concern as environmental contaminants due to evidence of high persistency, long-range transport, bioaccumulation potential and/or toxicity (Jian et al., 2017; Houde et al., 2011; Sunderland et al., 2019; Steenland et al., 2010; Barry et al., 2013; Ballesteros et al., 2017; Bach et al., 2015; de Voogt and Saez, 2006; Ghisi et al., 2019). PFOS and its related C₈ chemistry were phased out of production in North America during 2000–2002, banned in European countries, and listed in 2009 under the Stockholm Convention on Persistent Organic Pollutants as an Annex B substance (Buck et al., 2011; Paul et al., 2009). Perfluorooctane carboxylate (PFOA), its salts and PFOA-related compounds was listed under Annex A recently in May 2019. However, the number of known PFAS is much larger (Wang et al., 2013a). Buck et al. detailed 42 families and subfamilies and 268 individual PFAS which might be of environmental importance (Buck et al., 2011).

Thus far, only a small fraction of PFAS have been studied or evaluated from an environmental perspective (Buck et al., 2011), and most studies have focused on perfluorinated carboxylates (PFCAs), perfluorinated sulfonates (PFSAs) and related precursors, which have relatively low molecular weights (Ruan and Jiang, 2017). The production volumes of PFAS, most of which are likely precursors, greatly exceed those of PFCAs and PFSAs (Buck et al., 2011). There are at present a large number of PFAS other than PFACs and PFSAs that have not been accounted for in the environment. For example, measured PFCAs and PFSAs accounted only for 2–44 % of the anionic fraction of the extractable organic fluorine (EOF) in surface sediment from Lake Ontario in Canada, indicating that a large proportion of fluorine in this fraction was not characterized or quantified (Yeung et al., 2013).

Side-chain fluorinated polymer surfactants are important commercial PFAS products, and refer to those polymers with polyfluoroalkyl or perfluoroalkyl chains attached to non-fluorinated polymer molecular backbones (Buck et al., 2011; Russell et al., 2008; Washington et al., 2009; Chu and Letcher, 2014). Side-chain fluorinated polymers are extensively used for water-, stain- and grease-proofing finishes for textiles, leather products, and paper surfaces in numerous industrial and commercial applications (Buck et al., 2011; Hamid et al., 2018). It was reported that side-chain fluorinated polymers represent a high percentage of all PFAS-containing products, and more than 80 % in the case of fluorotelomers (Liu and Avendano, 2013).

Scotchgard is an important side-chain fluorinated polymer product marketed by the 3M Company since 1956 (Hamid et al., 2018). These side-chain fluorinated polymers (e.g., fluorinated acrylate-, methacrylate- and urethane-based polymers) are synthesized from key intermediates such as PFOS, perfluoroalkyl sulfonamides and perfluoroalkyl sulfonamide alcohols (Buck et al., 2011; Hamid et al., 2018). When they are released to the environment, abiotic and microbial degradation of non-fluorinated functionalities, and polyfluoroalkyl and perfluoroalkyl moieties will generate PFCAs or PFSAs (Chu and Letcher, 2014; Liu and Avendano, 2013). However, there are very few data available on the environmental fate and behavior of these side-chain fluorinated polymers, and their pollution sources, translation and degradation remain unclear.

PFAS can be released from consumer products during their lifespan and discharged into wastewater, thus entering wastewater treatment plants (WWTPs). Most WWTPs generate sludge and biosolids in either physical or biological wastewater treatment processes, but they are not

equipped or designed to remove some persistent pollutants such as PFAS, and thus can be pathways of PFAS to the environment (Arvaniti and Stasinakis, 2015; Hamid et al., 2018; Allred et al., 2015; Alder and van der Voet, 2015; Eggen et al., 2010; Gewurtz et al., 2013; Guerra et al., 2014; Lee et al., 2014; Navarro et al., 2018). High PFAS concentrations in sewage sludge and biosolids from WWTPs have been reported (Arvaniti and Stasinakis, 2015; Hamid et al., 2018; Gewurtz et al., 2013). Biosolids are nutrient- and energy-rich materials, and widely used as a fertilizer substitute and/or soil conditioner for carbon sequestration. In Canada more than 660,000 dry tonnes of biosolids are generated annually (Jin et al., 2018), and about 43 % of are used in land applications (Wang et al., 2013b). Therefore, there can be a risk of PFAS exposure to humans or wildlife from biosolids applications (Braunig et al., 2019).

We previously reported that the main side-chain fluorinated polymer components of Scotchgard fabric protector (pre-2002) and (post-2002) were detectable in 100 % and 80 %, respectively, in soil samples from biosolids-augmented agricultural sites in southern Ontario (Canada) (Chu and Letcher, 2017). The present hypothesis is that these side-chain fluorinated polymer components in the soil were derived from biosolids that were land-applied. To address this knowledge gap, the objectives of the present study are to 1) assess the occurrence of these Scotchgard derived side-chain fluorinated polymer surfactants in biosolids generated from WWTPs, and 2) examine the relationships between the locations and the type of WWTP processes and the levels of these side-chain polymer components in biosolids, and in comparison to the levels of other known PFAS.

2. Material and methods

2.1. Chemicals and materials

Scotchgard™ pre- and post-2002 formulations (Tech. mix) were acquired from AccuStandard Inc. (New Haven, CT, USA). These two standard solutions were based on commercial Scotchgard™ fabric protector products from the 3M Company (3M). These commercial products are comprised of solvents (acetone, isopropanol and liquefied petroleum gas), with the active compounds of fluorourethane (side-chain fluorinated polymer surfactants) making up only a few percent. Therefore, we evaporated away the solvents in the standard solutions supplied by AccuStandard and working standard solutions were prepared from the residual fluorourethane component.

According to the method of Benfodda et al. (2010), we generated a perfluoro-1-butane-sulfonamide (FBSA) standard. Perfluoro-1-ethylcyclohexyl-sulfonate (PF₁₁EtCH₂SO₃Na) was obtained from Campro Scientific GmbH (Berlin, Germany). Non-isotopically labelled and labelled PFAS standard solutions were procured from Wellington Laboratories (Guelph, ON, Canada). The full names and acronyms of the PFAS are listed in support information Table S1. PSA silica (polymerically bonded, ethylenediamine-*N*-propyl phase) was purchased from Sigma-Aldrich (Oakville, ON, Canada). Oasis WAX SPE cartridges (60 mg × 3 mL) were purchased from Waters Limited (Mississauga, ON, Canada). Ammonium hydroxide solution (28–30 %), Ultra CHROMASOLV grade methanol (MeOH), acetonitrile (ACN), hexane and dichloromethane (DCM) were purchased from Sigma-Aldrich. Ultrapure water was obtained from a Milli-Q system.

2.2. WWTP samples

Biosolids samples were collected from twenty WWTPs in eight provinces across Canada. Grab samples were collected using stainless steel pails and stored in pre-cleaned amber glass jars (Systems Plus, Baden, ON, Canada). Table 1 summarizes the codes, treatment methods, water flows, sampling years, and human populations served by these WWTPs. Names and locations of the WWTPs are not provided because the WWTPs participated on the condition of anonymity. The

Table 1

Wastewater treatment plant (WWTP) biosolids sampling locations from across Canada and related sampling site information.

WWTP No.	Plant Code	Process type ^a	Solids Treatment type ^b	Biosolids destination	Sampling year	Average flow m ³ /d	Population served (x1000 inhab.)
1	B	AT	AN	land application	2016	338,100	946
2	BV	ST	AN	land application	2017	22,900	50
3	C	ST	AN	land application	2017	16,900	36
4	CD	ST	AER	land application	2016	2,580	10
5	E	AT	DEW	Compost	2017	35,800	132
6	F	ST	DEW	Compost	2014	19,000	58
7	HG	ST*	AER	land application	2017	2,200	3
8	N	PT	AN	forest fertilization	2012	29,800	90
9	NP	ST	AN	land application	2017	5,000	15
10	PA	ST*	AER	land application	2017	4,000	12
11	PC	AT	AN	land application	2016	98,700	200
12	PD	ST	AN	land application	2017	38,700	70
13	PN	AT	DEW	Compost	2017	11,700	34
14	PW	ST	AN	land application	2017	35,100	63
15	Q	ST	AN	land application	2014	251,900	500
16	RM	ST#	PEL	land application	2016	160,000	234
17	SM	AT	ALK	land application	2017	10,200	15
18	U	PT	DEW	Incineration	2014	N/A ^c	N/A
19	V ^d	PT	ALK	land application	2017	91,400	350
20	W	ST	AN	land application	2016	202,900	705

^a AT (advanced, biological nutrient removal); ST (secondary, activated sludge); ST* (secondary, extended aeration); ST# (secondary, biological aerated filter); PT (primary, chemical addition).

^b DEW (dewatering); PEL (dewatering, pelletization); ALK (dewatering, alkaline stabilization); AN (digestion (mesophilic anaerobic)); AER (digestion (aerobic)).

^c N/A = not available.

^d A mixture biosolid sample from a few WWTPs.

WWTPs consisted of five advanced treatment (AT) plants, twelve secondary treatment (ST) plants, and three primary treatment with chemical addition (PT) plants. Advanced treatment (AT) included biological nutrient removal processes, which removes nitrogen and phosphorous in addition to oxygen demand through a series of different microbial environments. Secondary treatment (ST) included conventional activated sludge, extended aeration activated sludge, and biological aerated filter processes. Primary treatment (PT) included chemical addition to physical clarification to improve the overall solids removal. Different treatments were used to obtain biosolids. The rate of biosolids production in these plants ranged from 50 to > 15,000 dry total solid tonnes/year. Most sample collections (n = 16) was during 2016 and 2017, three samples were collected in 2014 and one sample was collected in 2012. The biosolids samples were kept frozen (−80 °C) until analysis.

2.3. Sample preparation and extraction for Scotchgard™ fabric protector (pre-2002 and post-2002) components and other PFAS

The biosolids samples were air-dried in a laboratory fume hood at room temperature. The air-dried biosolids samples were ground and sieved by a 60 mesh sieve. For Scotchgard™ fabric protector component analysis, about 0.2 g of sample was weighted and placed in a glass tube. Samples were extracted three times by adding 8 mL of 50/50 v/v acetone/hexane each time, vortexing and ultrasonication for 30 min at 45 °C (Symphony™, VWR, PA) followed by centrifugation (1800 g). All supernatants were combined and evaporated to dryness under a gentle stream of nitrogen. Residues were reconstituted in 2 mL ACN by vortexing and sonicating for 20 min at 45 °C. PSA silica sorbent (200 mg) was added to each reconstituted sample and mixed well by vortexing. The sample including the sorbent was transferred into a centrifuge filter with a modified nylon membrane (VWR, Mississauga, ON, Canada), which was pre-washed with ACN. After centrifugation the filtrate was transferred into a centrifuge tube and concentrated to 1 mL.

A standard addition method was used for the quantitative analysis of these side-chain fluorinated polymers. Among the individual samples the concentrations of the target compounds can be very different from each other, and thus a suitable concentration range for the standard addition must first be determined for the individual samples. To

estimate the concentration range of target compounds in sample, pre-analysis of the 1 mL reconstituted filtered samples on an UHPLC-MS/MS (see section 2.4) was performed with 1 µL injection and external standards method. Based on the pre-analysis, sample solution volumes were adjusted to get the concentrations of the target compounds in the samples to fall approximately within the 1–10 ng/mL concentration range in preparation for quantitative analysis. Four sub-sample solution fractions (with a volume of 100 µL) for each individual sample were prepared and then spiked individually with 100 µL ACN (concentration was 0) and three levels of standard solutions with comparable concentration of target compound in sample solution. The samples were then ready for UHPLC-MS/MS analysis.

The sample preparation and extraction method for all other 22 PFAS has been described elsewhere (Chu and Letcher, 2017; Chu et al., 2016) with some modifications and briefly described as follows. The sample was spiked with internal standards and ultrasonicated and extracted three times by 4 mL 0.2 % formic acid in ACN solution each time at 60 °C. The extracts were combined, concentrated to 2 mL and diluted by 8 mL water. The diluted extract was cleaned up and fractioned by Oasis WAX SPE cartridges (60 mg × 3 cc, Waters Limited, ON, Canada).

The moisture in the samples were measured gravimetrically by drying biosolid sub-sample in oven at 120 °C overnight. The concentration of target compounds in the sample was reported on a dry weight (d. w.) basis.

2.4. UHPLC-MS/MS and LC-Q-TOF analysis

The major components of the Scotchgard™ fabric protector products were determined in the biosolids sample extract and the analysis has been fully described elsewhere with some modifications (Chu and Letcher, 2017). An Agilent system comprised of a 6520A liquid chromatography-quadrupole time-of-flight mass spectrometer and Agilent 1200 LC (LC-Q-TOF-MS) (Agilent Technologies, Mississauga, ON, Canada) was used for the identification of standard solution components in the Scotchgard™ fabric protector products, as well as the confirmation of the results from UHPLC-MS/MS analysis of the biosolids samples. LC-Q-TOF-MS operational parameters were described in our previously published paper with some modifications (Chu and Letcher, 2014). A Kinetex® EVO C18 LC column (1.7 µm particle size, 100 Å,

2.1 × 50 mm, Phenomenex, CA, USA) was used. The mobile phase flow rate was 0.3 ml/min. The gradient started at 2 % B and held for 0.5 min, then increased to 95 % B in 4.5 min and held for 5 min; increased to 100 % B and held for 4 min, and then returned to initial conditions and the column was allowed to equilibrate for 10 min.

Quantitative analysis was performed on a Waters XEVO-TQ-S UHPLC-MS/MS (Waters Limited, Milford, MA, USA). The LC column was the same as that used in LC-Q-TOF-MS analysis as described above. Column and sample temperatures were 60 °C and 20 °C, respectively. The water (A) and methanol (B) mobile phases were buffered with ammonium acetate (2 mM). At a flow rate of 0.6 mL/min, the initial gradient was 2 % B (held for 0.5 min) and then increased to 95 % B over 2.5 min (held for 5 min). The composition of the mobile phase was then brought to the initial conditions including column equilibration for 6 min.

The XEVO-TQ-S UHPLC-QQQ-MS/MS was operated with an electrospray ionization source (ESI) and in the negative mode. The nebulizing gas and desolvation gas was nitrogen and the collision gas was argon. The capillary voltage was 1.5 KV. The desolvation temperature was 550 °C and the flow rate was 1000 L/h. The cone temperature was 150 °C and the flow rate was 150 L/h. The multiple reaction monitoring (MRM) mode was used for the analyses, and compound-dependent operation parameters and transitions have been described previously (Chu and Letcher, 2017).

Internal standards for the main components of Scotchgard™ fabric protectors (pre-2002) and (post-2002) (S1 and S2, respectively) were unavailable at the time of this study. Therefore, a standard addition method was used to determine S1 and S2 concentration and for the purpose of eliminating severe matrix effects. Four levels of S1 and S2 standard addition samples were prepared for each test sample and analyzed. A linear regression was made depending on spiked concentration of S1 and S2 in individual sub-samples and their response on MS/MS (Figure S1). The concentrations of S1 and S2 in the original sample solution could be calculated from the negative intercept on the x-axis.

Twenty-two other PFAS were also analyzed in the biosolid samples, which included thirteen perfluorinated carboxylic acids (PFCAs), five perfluorinated sulfonic acids (PFASs) and four perfluoroalkyl sulfonamides (FASAs) (Table 2 and Support Information Table S3). The UHPLC-MS/MS method of analysis for these other PFAS has been fully described in our previous publications (Chu and Letcher, 2017; Chu et al., 2016; Chu and Letcher, 2018).

2.5. Quality assurance and quality control

For each biosolids sample, four standard addition samples were analyzed and regression coefficients (r^2) higher than 0.99 were accepted (see Figure S1). Quality control standards, including duplicate samples and procedural blanks were added to each batch of samples ($n = 10$) and treated in the same way as the real samples. The method limits of detection (MLODs) and method limits of quantification (MLOQs) for S1 and S2 were defined as the concentration of target compounds in biosolids samples producing a peak in chromatogram with a signal-to-noise (S/N) ratio of equal to 3 and 10 (peak-to-peak), respectively. Recoveries of the target analytes were assessed by five replicate analysis of spiked biosolids samples.

3. Results and discussion

3.1. Determination of main components of Scotchgard™ fabric protectors in biosolids samples

There are two main components of Scotchgard™ fabric protectors (pre-2002 and post 2002) abbreviated as S1 and S2, respectively, which we have previously reported (Chu and Letcher, 2014). As we encountered with S1 and S2 in soil and sediment samples (Chu and Letcher, 2017), one of the major challenges associated with analysis of side-chain fluorinated polymers was extraction of the target compounds from the biosolids samples and subsequent clean-up for LC-MS analysis. For biosolids, different extraction methods were tested, which included accelerated solvent extraction (ASE) and sonication extraction with different solvents (such as acetone, ACN, toluene, and α,α,α -trifluorotoluene). Sonication with acetone/hexane (50/50 v/v) proved to be most ideal for S1 and S2 extraction. In the modified method, the sample extract was cleaned up further by dispersive solid phase extraction (d-SPE) using a PSA sorbent, which greatly improved S1 and S2 recoveries, and especially for S2. The target compounds could be detected by highly selective UHPLC-MS/MS with less interference. Biosolids samples are very complex matrices and matrix effects on MS/MS varied from 65 % to 241 % in these samples (calculated by the method reported by Matuszewski et al. (2003)). As shown by the UHPLC-MS/MS extracted ion chromatograms in Fig. 1, S1 and S2 could clearly be detected in biosolids samples, and their retention times matched those standards very well (as internal standard addition method was used). Regarding quantification, a typical sample analysis by UHPLC-MS/MS

Table 2

Summary of the concentrations of S1, S2 and some other PFAS as well as their fluorine content in biosolids samples from wastewater treatment plants (WWTPs) across Canada (see Table 1). (Unit: ng/g d.w.).

Code	S1	S2	PFBS	PFOS	FBSA	FOSA	Σ PFAS ₂₂	Σ F _{S1&S2}	Σ F _{PFAS22}
B	53.2	2051	0.32	11.6	0.10	1.90	43.4	228	32.1
BV	24.8	1035	< MLOQ	2.97	0.03	0.58	13.3	114	8.50
C	29.7	798	ND	5.70	0.02	0.97	19.4	90.8	13.2
CD	27.7	1239	3.48	19.0	0.67	1.84	58.4	136	43.0
E	17.8	1060	ND	8.90	0.05	0.97	25.2	115	19.1
F	22.2	349	ND	5.13	ND	0.31	13.0	41.9	8.24
HG	6.1	351	3.07	50.4	2.03	3.54	92.6	38.3	67.7
N	104.8	1824	< MLOQ	2.73	< MLOQ	0.81	13.6	217	8.52
NP	45.6	966	< MLOQ	3.40	0.02	0.99	14.4	112	9.08
PA	4.26	335	3.11	17.7	6.38	2.24	73.4	36.1	52.7
PC	42.7	1783	< MLOQ	14.2	0.09	5.37	67.9	197	52.9
PD	32.1	987	< MLOQ	13.1	0.02	1.38	29.0	111	19.5
PN	33.0	1770	ND	5.61	0.04	0.65	15.5	193	10.4
PW	20.0	894	0.29	16.5	0.04	1.03	33.2	98.5	22.3
Q	23.5	692	ND	17.9	0.02	1.27	31.2	78.2	22.0
RM	6.2	385	0.24	4.54	ND	0.26	12.1	41.8	6.00
SM	1.08	37.5	ND	0.49	ND	0.07	4.93	4.19	3.38
U	2.82	78.4	ND	2.32	< MLOQ	0.07	7.74	8.90	4.12
V	2.50	244	ND	2.87	< MLOQ	0.24	6.77	26.2	4.23
W	53.9	1094	1.58	4.97	< MLOQ	1.39	17.5	127	11.1

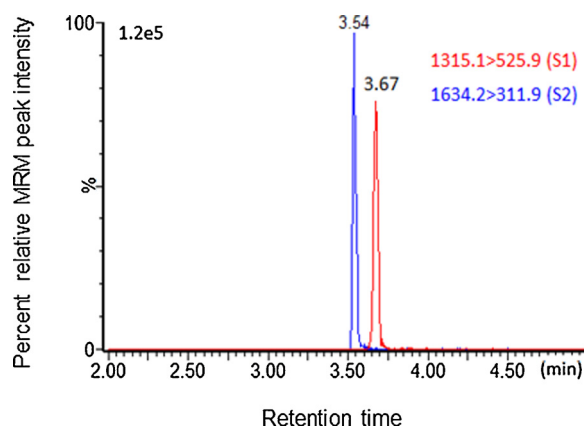


Fig. 1. UHPLC-MS/MS extracted ion chromatograms of main components (S1 and S2) in Scotchgard™ fabric protectors (pre-2002 and post-2002) in a representative biosolids sample from a WWTP.

using a standard addition method approach is shown in Figure S1. The recoveries, MLODs, MLOQs and ME for S1 and S2 were listed in Table S2. There were no detectable S1 and S2 in process blank samples.

To confirm the identity of S1 and S2 in the biosolid samples by UHPLC-MS/MS, LC-Q-TOF-MS (high resolution MS) was used to analyze some of the biosolids sample fractions, and the mass spectra were compared with the mass spectra for the main components of the Scotchgard™ fabric protector products (pre-2002 and post-2002)

standard solutions. Fig. 2 (a, b) showed product mass spectra with precursor ions of m/z 1315.0520 (S1) and m/z 1634.3015 (S2) in a representative biosolids sample. The product mass spectrum of m/z 1315.0520 (S1) showed two main ions of m/z 418.9737 and 525.9770 (Fig. 2a), which corresponded to fragment ions of $[C_8F_{17}]^-$ ($m/z = 418.9734$) and $[C_8F_{17}SO_2N(C_2H_5)]^-$ ($m/z = 525.9775$), respectively. The product mass spectrum of m/z 1634.30 (S2) showed two main ions with m/z 218.9855 and 311.9746 (Fig. 2b), which corresponded to the fragment ions of $[C_4F_9]^-$ ($m/z = 218.9856$) and $[C_4F_9SO_2N(CH_3)]^-$ ($m/z = 311.9746$), respectively. The full composition of Scotchgard™ fabric protector products are a trade secret (3M Scotchgard, 2018). The only known characteristic molecular structures of S1 and S2 are illustrated in Fig. 2. The commercial Scotchgard™ fabric protector safety data sheets (3M Scotchgard, 2018) state that the polymer backbone (Y and Y' for S1 and S2 in Fig. 2) is urethane. Although it was reported that the chemical group of $-NHC(O)OCH_2CH_2-$ is a commonly used hydrocarbon chain (X and X' for S1 and S2 in Fig. 2) for creating fluorotelomer-based urethane polymers (Buck et al., 2011), the MS data could indicate but not fully confirm the urethane polymer backbone and the hydrocarbon chain in the case for the S1 and S2 molecular structures in this present study.

3.2. Side-chain fluorinated polymer surfactants (S1 and S2) in biosolid samples from WWTPs

High concentrations of S1 and S2 were measured in all biosolids samples from WWTPs from across Canada. S1 concentrations ranged

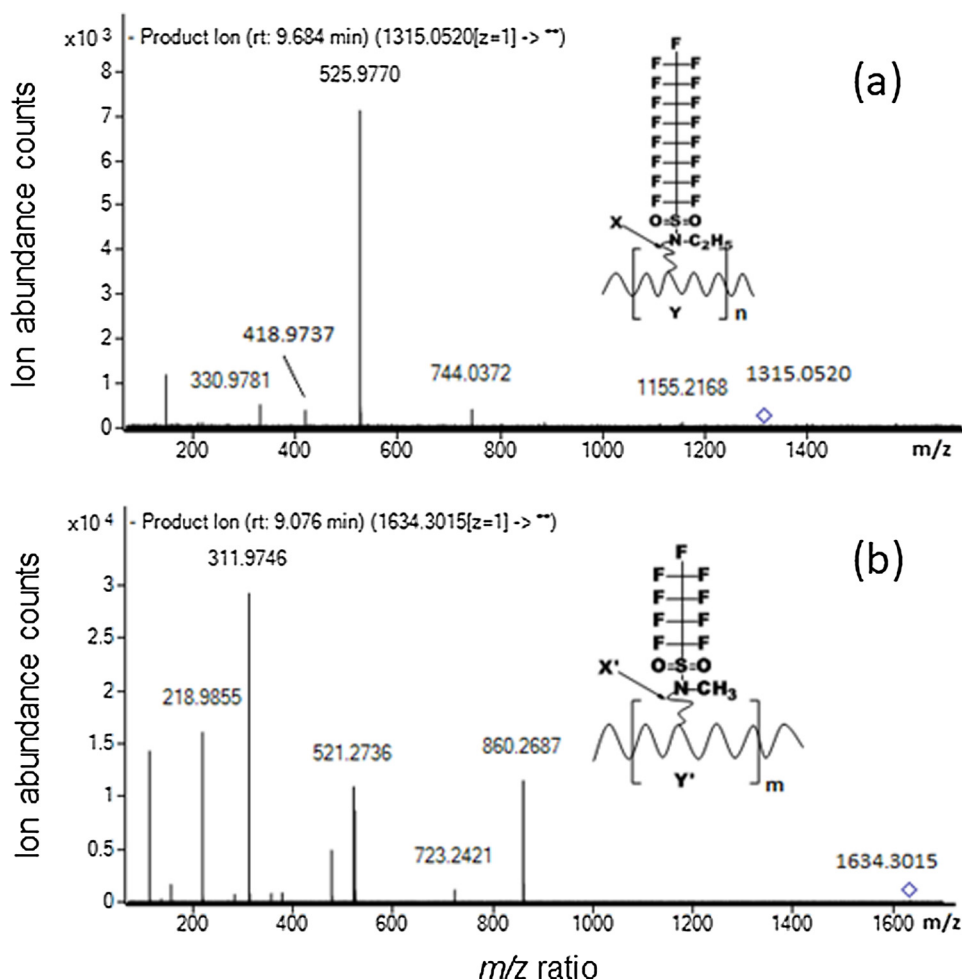


Fig. 2. LC-Q-ToF-MS generated mass spectra of precursor ions m/z 1315.0520 (a; S1) and m/z 1634.3015 (b; S2) at retention times of 9.7 and 9.1 min in the mass chromatogram of a representative biosolids sample from a WWTP.

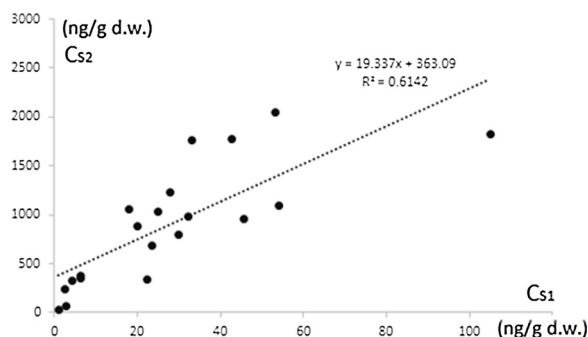


Fig. 3. Regression analyses between S1 and S2 concentrations in the WWTP biosolids samples.

from 1.08–105 ng/g d.w. were consistently lower than S2 concentrations which ranged from 37.5–2051 ng/g d.w. (Table 2). The considerably higher S2 concentrations in the 2012–2017 biosolids were due to the heavier use of S2 in the Scotchgard™ fabric protectors post-2002 although the S1 that dominated pre-2002 products was still clearly being used during the 2012–2017 sampling period. A strong and significant correlation between S1 and S2 concentrations in the biosolids samples ($p < 0.001$; $r^2 = 0.6142$) indicated that S1 and S2 originated from similar sources (Fig. 3). Compared to all other PFAS (Table 2), the sum concentrations of S1 and S2 were approximately thirty times higher than total concentration of all other PFAS in the samples.

Although the biosolids samples came from WWTPs, which employed a wide range of treatment processes (detailed in Table 1), no correlations were apparent between different liquid or solid WWTP processes and the observed for S1 (or S2) concentrations (Table 2). Also, no significant correlations ($p > 0.05$) were found between S1 (or S2) concentrations and the human population served or the average flow rate of the WWTPs. However, there was an inverse correlation ($p < 0.05$) between concentrations of S1 (or S2) with the ratio of average flow rate in the WWTPs to the human population served ($\text{m}^3/\text{person/day}$) (Fig. 4). This indicated that S1 and S2 mainly comes from domestic wastewater, and that other wastewater sources coming into the WWTPs resulted in dilution of these PFAS.

3.3. Occurrence of other PFAS in biosolids samples

The mean of the sum concentration of the 22 other PFAS (ΣPFAS_{22}) in the biosolids samples ranged from 4.93 to 92.6 ng/g d.w. (Tables 2 and S3). The concentration range of ΣPFAS_{22} is comparable with the PFAS results from a previous study for WWTPs across Canada (Guerra et al., 2014). The present PFAS concentrations (with exclusion of S1 and S2) in biosolids are also comparable with other studies on PFAS in sludge or biosolids samples from municipal WWTPs, with exception of the samples from the locations near PFAS manufacturers (Arvaniti and Stasinakis, 2015). For the present biosolids samples collected from 2012 to 2017, PFOS, PFDA and PFOA were dominant of the 22 PFAS. Although after 2002 manufacturers were using more shorter-chain PFAS with < 7 fluorinated carbons, only in 30 % and 60 % of the present samples were PFBA and PFBS detected from sample collected during 2012–2017. This is due to these shorter-chain fluorinated products having relatively lower potential to adsorb onto or absorb into the biosolids matrices (Arvaniti and Stasinakis, 2015; Sinclair and Kannan, 2006). In all the present samples, FOSA was measureable at a low concentration range of 0.07–5.37 ng/g d.w. FBSA was detected in 85 % of the samples with a comparably low concentration range of < 0.01 to 6.38 ng/g d.w. Samples from the HG and PA WWTPs (Table 1) contained higher concentrations of FBSA, FOSA and some other PFAS with relatively shorter-chains than that in the other WWTP samples. The HG and PA samples came from WWTPs with process type of extended

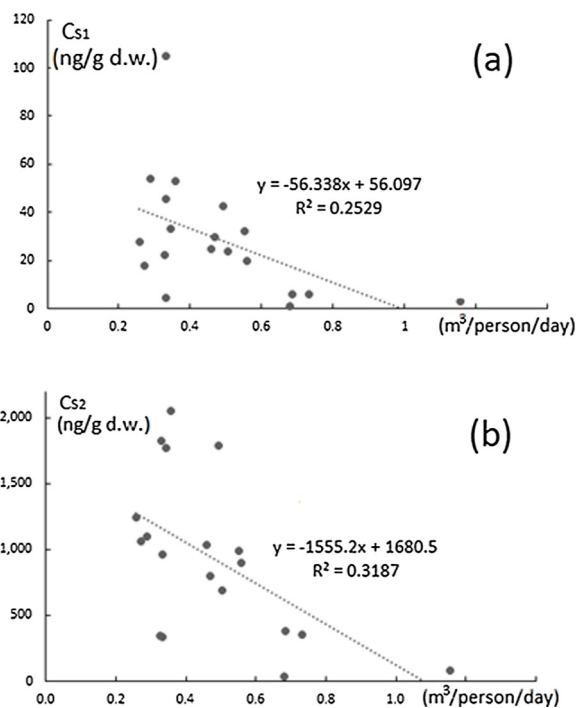


Fig. 4. The relationship between concentration of S1 (or S2) with the ratio of the average flow rate in the WWTPs to the served human population ($\text{m}^3/\text{person/day}$).

aeration activated sludge and aerobic digestion. It may be that if longer-chain PFCAs or PFASs are degraded under aerobic conditions it would happen to the greatest extent at the HG and PA WWTPs. No significant ($p > 0.05$) correlations were observed between concentrations of S1 (or S2) and any of these other 22 PFAS (Table 3). This indicated that these PFAS have complex pollution sources and environmental behaviour that warrant further study. Other sources, such as road runoffs and industrial discharges, also likely contribute to these 22 PFAS in the present biosolids samples.

Even-odd pairs have been shown to be a useful metric to examine the sources of PFCAs (MacInnis et al., 2017; Liu et al., 2011). For short-chain PFAS, a pattern where the concentration of even-carbon numbered PFCAs was relatively higher than that of odd-carbon numbered PFCAs was observed in the present biosolid samples (Table S3). Similar PFCA patterns were also observed in landfill leachates from 22 sites in Germany (Busch et al., 2010) and WWTP sewage sludge from Switzerland (Alder and van der Voet, 2015). A correlation analysis was performed in the present study and the results showed that for short-chain PFCAs the correlation coefficient (r) values between even-carbon numbered PFCAs to even-carbon numbered PFCAs (such as PFHxA to PFOA, or PFHxA to PFDA) tended to be higher than that between even-carbon numbered PFCAs to odd-carbon numbered PFCAs (such as PFHxA to PFNA, or PFHxA to PFUnDA) (Table 3). PFBA, PFPeA and PFHpA were not considered due to their low detected frequency in the samples. For the long chain PFCAs (such as PFDoDA to PFTeDA, or PFDoDA to PFTeDA) with odd- or even-carbon numbers, the correlation coefficient (r) values were similar (Table 3). This indicated that in these samples the short-chain PFCAs mainly come from commercial products manufactured by fluorotelomer iodide oxidation technology but the long-chain PFCAs mainly come from commercial products manufactured by electrochemical fluorination technology (ECF) (Buck et al., 2011; Prevedouros et al., 2006).

Table 3

Correlation analysis results for perfluorinated carboxylic acids (PFCAs) measured in biosolids samples from wastewater treatment plants across Canada.

PFCAs	PFHxA	PFOA	PFNA	PFDA	PFUdA	PFDoA	PFTTrDA	PFTeDA	PFHxDA	PFODA
PFHxA	1.0000									
PFOA	0.7100	1.0000								
PFNA	0.2229	0.3306	1.0000							
PFDA	0.5605	0.6871	0.7916	1.0000						
PFUdA	0.1735	0.1324	0.8452	0.7764	1.0000					
PFDoA	0.3675	0.3363	0.6375	0.7766	0.8153	1.0000				
PFTTrDA	−0.0406	−0.1669	0.3956	0.2804	0.6453	0.7189	1.0000			
PFTeDA	0.5858	0.5960	0.5057	0.7969	0.6543	0.8538	0.5649	1.0000		
PFHxDA	0.3802	0.2558	0.4237	0.5656	0.6492	0.7848	0.7268	0.8702	1.0000	
PFODA	−0.1083	−0.4512	−0.0929	−0.1530	0.1074	0.1155	0.3905	−0.0146	0.3194	1.0000

3.4. Organic fluorine content in S1, S2 and other PFAS in biosolids from WWTPs

In addition to commonly monitored PFAS, a larger number of other PFAS are likely to be present in environmental samples, but thus far have eluded direct detection using common analysis methods (Yeung et al., 2013; Liu and Avendano, 2013; Sinclair and Kannan, 2006; Becker et al., 2008; Murakami et al., 2009; Schultz et al., 2006; Codling et al., 2015). Codling et al. (2015) measured extractable organic fluorine (EOF) and PFAS in dated sediment cores and grab surface sediments from Lake Michigan. The total elemental fluorine in quantified PFAS accounted for only 0.16 % and 0.09 % of total EOF in sediment cores and grab sediment samples, respectively (Codling et al., 2015). Yeung et al. (2013) reported that worms and gastropods, organisms occupying lower trophic levels and living in the sediment, contained the greatest amount of unidentified organic fluorine compounds compared to fish that occupied higher trophic positions. The suggestion was made that significant amounts of unidentified organic fluorine compounds were present in sediment and could be the precursor and/or intermediate compounds of PFASs and PFCAs (Yeung et al., 2013). A large number of customer products were analyzed for 39 PFAS, as well as EOF, and the measured 39 PFAS concentrations could only explain a small fraction of the EOF in the samples (Schultes et al., 2018).

In the present study, elemental fluorine in the main component of side-chain fluorinated polymer surfactants (S1 and S2) and other 22 PFAS were calculated depending on their molecular structures (Table 2). In the calculation, we assumed that in the S1 or S2 molecular structures there were only one side-chain fluorinated branch for each. Even so, in most of these biosolid samples the concentrations of elemental fluorine in S1 and S2 ($\Sigma F_{S1\&S2}$) were still much higher than that in the total concentration of other 22 PFAS (ΣF_{PFAS22}). The ratio of $\Sigma F_{S1\&S2}$ vs. ΣF_{PFAS22} ranged from 0.6–25. This can explain the reason, at least partially, why in some sludge or sediment samples the common measured PFAS could only explained a small fraction of their EOF (Codling et al., 2015).

Given the high production and wide usage in commercial products, the environmental presence and behavior of side-chain fluorinated polymers environmental remains largely unknown. It remains ambiguous if these high molecular weight side-chain fluorinated polymers are precursors and significantly contribute to PFAS in environment (Liu and Avendano, 2013). Moreover, these side-chain fluorinated polymers can encompass many other potential structures and products (Buck et al., 2011).

4. Conclusions

Based on our mass spectrometric analysis findings, S1 and S2 can be chemically described as side-chain perfluorooctane sulfonamide-urethane polymer and side-chain perfluorobutane sulfonamide-urethane polymer, respectively. The present research showed that the main S1 and S2 components in Scotchgard™ fabric protector products are

present in biosolids from pan-Canadian wastewater treatment plants (WWTPs), and at comparable levels in the case of S1 and S2 but orders of magnitude greater concentrations relative to the sum concentrations of 22 other PFAS, which are mainly PFCAs and PFASs. PFAS concentrations in biosolids from WWTPs are likely to be highly underestimated without consideration of concentrations of S1 and S2 as well as other potential side-chain fluorinated polymer surfactants. Furthermore, there is the potential concern that these side-chain fluorinated polymers could environmentally degrade to generate bioavailable and bioaccumulative polyfluorinated sulfonamides such as FBSA and FOSA, which could subsequently degrade to e.g. PFBS and PFOS. Even if these are slow degradation processes, the high concentrations of S1 and especially S2 in WWTP biosolids may suggest a serious PFAS issue in the environment. Our results provide evidence that commonly monitored PFAS only explain a small fraction of extractable organic fluorine (EOF) in these biosolid samples. Clearly, more studies are warranted on (side-chain) fluorinated polymer surfactants in the environment.

Declaration of interests

Nothing declared.

CRediT authorship contribution statement

Robert J. Letcher: Conceptualization, Funding acquisition, Project administration, Supervision, Data curation, Investigation, Resources, Writing - original draft, Writing - review & editing. **Shaogang Chu:** Conceptualization, Investigation, Methodology, Data curation, Validation, Writing - original draft. **Shirley-Anne Smyth:** Methodology, Resources, Writing - original draft.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jhazmat.2020.122044>.

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